# Predicting Student Performance: A Dual Approach Using Classification and Regression Models

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***Abstract*—Accurately predicting student performance is essential for identifying at-risk learners and informing data-driven academic interventions. This paper presents a dual approach by analyzing two datasets: one formulated as a classification problem (student grade categories), and another as a regression task (continuous performance scores). Through an ensemble of machine learning algorithms—including Support Vector Machines, Random Forest, Gradient Boosting, and AdaBoost—we demonstrate that SVM achieves the highest accuracy (~75%) on discrete grades, while Random Forest Regressor yields an $R^2$ of 0.90 for continuous performance. A detailed correlation analysis reveals strong relationships between absences and grade class (0.73), and between prior GPA and final performance (0.93). These results underscore the importance of robust feature preprocessing, hyperparameter tuning, and the potential for automated early warning systems in educational contexts.**

***Index Terms*—Student Performance, Educational Data Mining, Classification, Regression, Ensemble Methods**

## I. Introduction

Accurately estimating student performance is a central challenge in modern education. With increasingly diverse student populations and rapid curriculum changes, educators and administrators need data-driven tools to detect at-risk learners early. However, the process of evaluating student performance can be complicated by biases or incomplete data—paralleling the feedback loops observed in other domains [1], [2].

To address these concerns, data mining techniques and machine learning (ML) methods have seen growing adoption in educational data mining (EDM). By leveraging both classification (for discrete grade categories) and regression (for continuous performance scores), we can approximate a "true" performance metric and reduce subjective bias [3], [4]. This project aims to highlight which features most strongly correlate with learning outcomes (e.g., absences, prior GPA, extracurricular activities), potentially guiding more targeted interventions and resource allocation.

Another goal is to compare the effectiveness of various ML algorithms—ranging from classic models such as Logistic Regression and Linear Regression to ensemble-based methods like Random Forest, Gradient Boosting, and AdaBoost. We apply these to two datasets:

- **Dataset1**: A classification-oriented dataset containing demographic, academic, and behavioral features with *GradeClass* as the target.
- **Dataset2**: A regression-focused dataset capturing continuous performance metrics (*Overall*), along with high school scores, attendance, and other personal factors.

By blending both classification and regression frameworks, this research showcases how student performance can be modeled at multiple levels of granularity. Our findings suggest that these approaches can help educators detect early-warning signs of dropout or academic failure and design tailored interventions that address students' unique needs.

## II. Research Objectives

1) **Classification vs. Regression:** Evaluate performance across standard classification models (SVM, Random Forest, etc.) for discrete grade classes (Dataset1), and regression models (Linear Regression, Gradient Boosting, etc.) for continuous performance (Dataset2).
2) **Correlation Analysis:** Identify the strongest correlating features in each dataset—such as *Absences* (Dataset1) and *Last* (Dataset2).
3) **Hyperparameter Tuning:** Apply GridSearchCV to refine model parameters and measure performance improvements.
4) **Implications for Educational Practice:** Demonstrate how combined classification/regression modeling can inform early interventions and resource distribution in learning environments.

## III. Literature Review

The quest to predict student performance has grown into a robust field known as *educational data mining (EDM)*, where ML techniques are leveraged to uncover patterns and risk factors [5], [6]. Early efforts often used basic statistical methods, but as student data evolved

(demographic, behavioral, textual), advanced ML emerged [7], [8].

### A. Evolving Paradigm of Predictive Analytics in Education

Structured student information (e.g., attendance, GPA) and unstructured sources (discussion forums, online logs) have expanded the predictive scope [9], [10]. Similar to eWOM feedback loops, biases may arise if initial low performance triggers lower expectations or less support, reinforcing negative outcomes [11]. Real-time monitoring and intervention systems have thus become pivotal [12], [13].

### B. ML for Predicting Student Performance

Ensemble methods—*Random Forest*, *Gradient Boosting*, *XGBoost*, *CatBoost*—often outperform single estimators due to their ability to capture non-linear feature interactions [14]–[16]. Hyperparameter tuning via cross-validation further improves these models [17]. *Support Vector Machines (SVM)* excel on moderate datasets with margin-maximization strategies [18], [19].

*Logistic Regression* remains a straightforward baseline for classification tasks [20], [21], while *Naive Bayes* and *K-Nearest Neighbors* can be competitive depending on data characteristics [22], [23].

### C. Feature Engineering and Key Predictors

As found in many studies, *Absences* frequently correlate strongly with final grades [24], reinforcing the notion that consistent engagement is critical. *StudyTimeWeekly*, *ParentalSupport*, or *Extracurricular* activities show moderate effects [25], [26]. For regression-based outcomes, *Last semester's GPA* or *SSC/HSC scores* are typically top predictors [27], [28].

Text-based sentiment analysis can also yield insights into a student's motivation or frustration. Techniques akin to word embeddings or transformer encoders have found success in extracting meaning from discussion boards or written reflections [29], [30].

### D. Classification vs. Regression Approaches

While many past works treat student performance as discrete categories (pass/fail or letter grades), regression frameworks offer finer resolution [31], [32]. Ensemble regressors (*Random Forest Regressor*, *Gradient Boosting Regressor*) often capture complex relationships better than linear models [33]. Tuning learning rates, depth, and n_estimators helps mitigate overfitting [34].

### E. Synthesis and Research Gaps

The literature consistently highlights the benefits of combining *demographic*, *behavioral*, and *historical academic* data [35], [36]. Yet, gaps remain around dynamic or longitudinal modeling, especially integrating real-time analytics from learning management systems. Transfer learning across institutions or courses is another under-explored frontier [37], [38].

**Conclusion:** Overall, advanced ML methods—particularly ensemble models—prove highly effective in predicting student outcomes. Addressing data imbalance, capturing real-time signals, and expanding feature spaces (e.g., textual feedback) are key directions for continued research.

## IV. Approach Methodology

Since our project aims to predict student performance in two different ways—**classification** (Dataset1) and **regression** (Dataset2)—we adopt a dual pipeline. This section describes the data preprocessing stages, the machine learning frameworks employed, and an overview of the datasets.

### A. Pre-processing

The pre-processing step consists of two parts: one for *Dataset1 (Classification)* and another for *Dataset2 (Regression)*. Both pipelines share fundamental tasks like handling missing values, encoding categorical variables, and scaling numeric columns.

*a) Dataset1 (Classification) Pre-processing:*

- **Data Cleaning:** We dropped any irrelevant columns (e.g., *StudentID*, *GPA*), confirmed minimal null values, and removed them if necessary.
- **Encoding Categorical Features:** Categorical columns (like *Gender*, *Ethnicity*, *ParentalSupport*) were label-encoded or one-hot encoded, depending on feature cardinality.
- **Scaling Numerical Columns:** Numeric columns (e.g., *Age*, *StudyTimeWeekly*, *Absences*) were standardized (mean=0, std=1) to avoid scale discrepancies across features.
- **Correlation Analysis:** A Pearson correlation heatmap identified that *Absences* had the strongest correlation (0.73) with *GradeClass*.

*b) Dataset2 (Regression) Pre-processing:*

- **Data Cleaning:** We ensured no extraneous columns or missing data. String-based fields (e.g., *Income* category) were label-encoded into ordinal/numeric form.
- **Scaling and Encoding:** Numeric features (like *HSC*, *SSC*, *Last*) were standardized via `StandardScaler`. Categorical attributes (*Gender*, *Hometown*, *Attendance* range) were label- or one-hot encoded.
- **Correlation Matrix:** *Last* (prior GPA) was found to be most predictive of *Overall* with correlation 0.93.

### B. Classification

The pre-processed *Dataset1* was then split into training (80%) and testing (20%). A suite of classification algorithms were employed:

- **Logistic Regression, K-Nearest Neighbors, SVM**
- **Decision Tree, Random Forest, Gradient Boosting, AdaBoost**
- **Gaussian Naive Bayes, XGBoost, CatBoost**

Model performance was evaluated using **accuracy** and confusion matrices. Hyperparameter tuning (via `GridSearchCV`) was optionally applied to top models (e.g., SVM, Random Forest). The best accuracy obtained was $\sim 0.75$ (SVM).

### C. Regression

The *Dataset2* pipeline focused on predicting a continuous performance metric (*Overall*). Four regressors were tested:

- **Linear Regression**
- **Random Forest Regressor**
- **Gradient Boosting Regressor**
- **AdaBoost Regressor**

We used an 80%-20% train-test split, computing **Mean Squared Error (MSE)** and $R^2$ to gauge performance. Random Forest Regressor emerged superior with an $R^2 \approx 0.90$ and MSE of 0.11. We further refined its hyperparameters (e.g., *n_estimators*, *max_depth*) using `GridSearchCV`.

### D. Datasets

- **Dataset1 (Classification):** Contained 2,392 rows with 15 features plus a categorical target *GradeClass* {*A,B,C,D,E*}. Key features included *Age*, *Gender*, *Absences*, *ParentalSupport*, *StudyTimeWeekly*.
- **Dataset2 (Regression):** Consisted of 493 rows and 16 features, with a continuous target *Overall*. Numeric columns like *HSC*, *SSC*, and *Last* (prior GPA) strongly influenced performance.

**Summary:** Both datasets followed similar data-cleaning and encoding practices, while each addressed different predictive goals (discrete vs. continuous). Classification metrics (accuracy/confusion matrix) were used for *Dataset1*, and regression metrics (MSE/$R^2$) for *Dataset2*. Notably, *Absences* and *Last* (prior GPA) emerged as top predictors in their respective tasks.

## V. Machine Learning Methods

In this study, we explored a total of **nine classification** algorithms and **four regression** algorithms to predict student performance. This section is organized into two main parts: Classification Models (Dataset1) and Regression Models (Dataset2).

### A. Classification Models

**Dataset1** involves discrete grade classes {A, B, C, D, E}. The following nine classifiers were employed:

*1) AdaBoost Classifier:* **AdaBoost** (Adaptive Boosting) reweights misclassified samples after each iteration, producing a weighted sum of weak learners:

$$\hat{y} = \operatorname{sign}\Big(\sum_{m=1}^{M} \alpha_m \, h_m(x)\Big)$$

where $\alpha_m$ depends on the error rate of each weak learner (e.g., short decision trees). AdaBoost is effective when combining many weak but diverse models.

*2) CatBoost Classifier:* **CatBoost** specializes in handling categorical features efficiently by employing permutation-driven encoding. Like other boosting approaches, CatBoost iteratively corrects errors of previous learners, but includes robust handling of non-numeric columns and balanced tree growth strategies.

*3) Decision Tree Classifier:* A **Decision Tree** recursively splits the feature space based on metrics such as Gini impurity or entropy. Each leaf represents a class prediction. While easy to interpret, single trees can overfit without pruning or ensemble strategies.

*4) Gradient Boosting Classifier:* **Gradient Boosting** constructs an additive ensemble of weak learners (often shallow trees) to model residual errors:

$$F_m(x) = F_{m-1}(x) + \nu \cdot h_m(x)$$

where $\nu$ is the learning rate. By sequentially correcting residuals, Gradient Boosting can attain high predictive accuracy if carefully tuned.

*5) K-Nearest Neighbors (KNN):* **KNN** is a non-parametric method storing the entire training data. A new sample is classified by majority vote of its $k$ nearest neighbors in feature space. Performance depends heavily on distance metrics and the choice of $k$.

*6) Logistic Regression:* **Logistic Regression** learns a linear decision boundary for classification. The probability of the positive class is given by the sigmoid function:

$$\sigma(z) = \frac{1}{1+e^{-z}}, \quad z = w \cdot x + b$$

Parameters $w$ and $b$ are found by minimizing cross-entropy loss.

*7) Random Forest Classifier:* **Random Forest** is an ensemble of decision trees. Each tree is trained on a bootstrap sample, and final predictions derive from majority vote:

$$\hat{y} = \operatorname{mode}\big\{h_1(x), \ldots, h_n(x)\big\}$$

This aggregation reduces variance and resists overfitting, making Random Forest robust for classification tasks.

*8) Support Vector Machine (SVM):* **SVM** aims to find a hyperplane that maximizes the margin between classes. In the linear case:

$$\min_{w,b} \frac{1}{2}\|w\|^2 \quad \text{subject to} \quad y_i(w \cdot x_i + b) \geq 1, \forall i$$

Non-linear datasets can be handled by kernel functions (RBF, polynomial), transforming the feature space.

*9) XGBoost Classifier:* **XGBoost** is a high-speed, regularized gradient boosting library. It updates an ensemble model in stages:

$$F_m(x) = F_{m-1}(x) + \eta \cdot h_m(x)$$

where $\eta$ (learning rate) regulates each new tree's contribution. Its second-order gradient approximations and advanced regularization often yield top performance in tabular data tasks.

### B. Regression Models

**Dataset2** focuses on predicting a continuous performance score (*Overall*). Four regressors were employed:

*1) AdaBoost Regressor:* **AdaBoost Regressor** extends Adaptive Boosting to continuous targets. Each weak learner fits current residuals, and the final prediction is a weighted sum of all learners:

$$\hat{y}(x) = \sum_{m=1}^{M} \alpha_m \, h_m(x)$$

This approach can reduce bias if base learners are suitably chosen.

*2) Gradient Boosting Regressor:* **Gradient Boosting Regressor** iteratively refines an ensemble of weak regressors to model residual errors:

$$F_m(x) = F_{m-1}(x) + \nu \cdot h_m(x)$$

Hyperparameters like learning rate, max depth, and number of estimators require careful tuning to avoid overfitting.

*3) Linear Regression:* **Linear Regression** is the simplest approach for modeling numeric outcomes:

$$\hat{y} = \beta_0 + \sum_{i=1}^{n} \beta_i x_i$$

These coefficients $\{\beta_i\}$ are typically estimated by minimizing Mean Squared Error (MSE).

*4) Random Forest Regressor:* **Random Forest Regressor** applies the Random Forest concept to continuous targets. Each decision tree outputs a real value, and the forest's overall prediction is the average:

$$\hat{y} = \frac{1}{n} \sum_{i=1}^{n} \hat{y}_i$$

This ensemble approach reduces variance and improves predictive stability.

**Summary:** These classification (AdaBoost, CatBoost, Decision Tree, Gradient Boosting, KNN, Logistic Regression, Random Forest, SVM, XGBoost) and regression (AdaBoost Regressor, Gradient Boosting Regressor, Linear Regression, Random Forest Regressor) algorithms cover a wide range of machine learning paradigms, allowing for both discrete (grade-class) and continuous (overall performance) student predictions.

## VI. Results and Discussion

In this section, we present and interpret our findings from both **Classification** (Dataset1) and **Regression** (Dataset2) tasks.

### A. Evaluation Metrics

*a) Classification Metrics::* To compare the nine classification models (Decision Tree, Random Forest, Gradient Boosting, KNN, SVM, Logistic Regression, XGBoost, CatBoost, AdaBoost), we focused on:

- **Accuracy** $= \dfrac{\text{Number of Correct Predictions}}{\text{Total Samples}}$
- **Confusion Matrix** for class-specific breakdown
- **F1-Score**, if class imbalance needed more attention

*b) Regression Metrics::* For the four regressors (Linear Regression, Random Forest Regressor, Gradient Boosting Regressor, AdaBoost Regressor), we used:

- **Mean Squared Error (MSE):**

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2$$

- **Coefficient of Determination ($R^2$):** a measure of how much variance in the target is explained by the model.

### B. Results

We break down our results for classification (Dataset1) and regression (Dataset2), highlighting key observations from **Figures** such as dataset distributions, correlation heatmaps, and final model performance charts.

*1) Classification Results (Dataset1)::*

- **Model Accuracies:** Table I compares the accuracies for nine classifiers. **Support Vector Machine (SVM)** achieved the highest accuracy at approximately 0.75, followed by Logistic Regression (0.73) and Random Forest (0.70).
- **Confusion Matrix:** (Figure 1) indicates that majority class "E" was predicted more accurately than the minority classes. This reflects class imbalance, with "E" representing about 50% of the dataset.
- **Discussion:** The top performing SVM suggests margin-based approaches adapt well to the dataset's features. Ensemble methods (Random Forest, Gradient Boosting) performed steadily but scored slightly below SVM. KNN, Decision Tree, and Naive Bayes showed lower accuracies, potentially due to overfitting or sensitivity to data distribution.
- **Comparison with Other Results:** To contextualize our findings, we compared our classification outcomes with prior works on the same dataset (as illustrated in Table II).

*2) Regression Results (Dataset2):*

- **Model $R^2$ Scores:** As illustrated in Table III, **Random Forest Regressor** emerged best ($R^2 = 0.90$, MSE=0.11). Linear Regression (0.89) and Gradient Boosting (0.89) closely followed, while AdaBoost scored around 0.86.
- **Correlation Heatmap:** (Figure 2) showed *Last* (prior GPA) had the strongest correlation (0.93) with *Overall*, suggesting consistent academic performance

is a key predictor. *HSC* and *SSC* also correlated moderately.

- **Discussion:** The strong result from Random Forest Regressor supports the idea that ensemble methods capture non-linearities in the data. Linear Regression performed surprisingly well for a simpler model, likely due to relatively linear relationships among numeric features.

TABLE I
SUMMARY OF CLASSIFICATION RESULTS

| Model | Accuracy |
|---|---|
| Logistic Regression | 0.73 |
| K-Nearest Neighbors | 0.65 |
| Support Vector Machine | 0.75 |
| Decision Tree | 0.57 |
| Random Forest | 0.70 |
| Gradient Boosting | 0.69 |
| AdaBoost | 0.69 |
| Gaussian Naive Bayes | 0.68 |
| XGBoost | 0.67 |
| CatBoost | 0.70 |

TABLE II
COMPARISON OF CLASSIFICATION MODELS

| | SVM | RF | GB | LR |
|---|---|---|---|---|
| Results | 0.75 | 0.70 | 0.69 | 0.73 |
| Swarnabh Das [39] | 0.63 | 0.67 | 0.69 | 0.58 |
| Vuong Anh Binh [40] | 0.62 | 0.70 | 0.70 | 0.61 |

TABLE III
SUMMARY OF REGRESSION RESULTS

| Model | MSE | $R^2$ |
|---|---|---|
| Linear Regression | 0.12 | 0.89 |
| Random Forest Regressor | 0.11 | 0.90 |
| Gradient Boosting Regressor | 0.12 | 0.89 |
| AdaBoost Regressor | 0.15 | 0.86 |

### C. Combined Discussion

**Classification vs. Regression Findings:**

- **Absences** emerged as a significant factor in classification tasks. Meanwhile, **Last** (prior GPA) dominated in regression, reflecting consistent academic trajectories.
- The synergy between robust *feature preprocessing* (label encoding, scaling) and ensemble ML approaches (Random Forest, Gradient Boosting) is evident in both tasks.
- For classification, SVM's margin-based optimization overcame class imbalance better than single trees. For continuous scores, ensemble methods captured variance more effectively than linear baselines.

Overall, the **Results** underscore the importance of **feature engineering** (e.g., handling categorical variables, scaling numeric columns) and **hyperparameter tuning** in achieving strong predictive performance. By comparing multiple algorithms, we validated that no single model universally dominates—*SVM* excelled at discrete grade prediction, while *Random Forest Regressor* led continuous outcome forecasts.

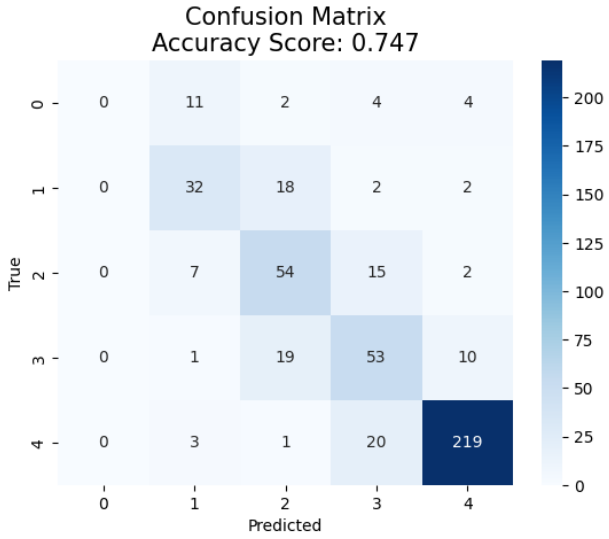


Fig. 1. Confusion Matrix for the best model (SVM) in Dataset1.

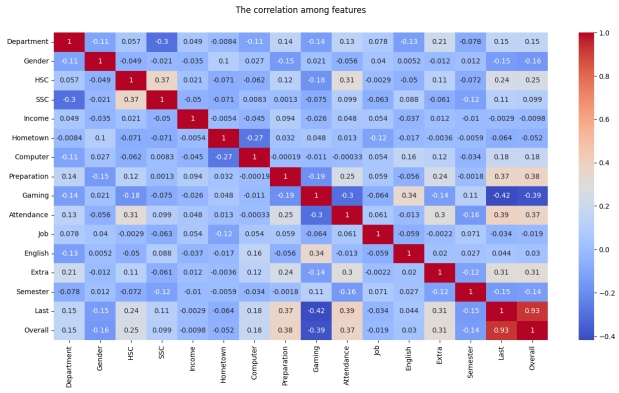


Fig. 2. Correlation Heatmap for Dataset2, showing *Last* correlated (0.93) with *Overall*.

## VII. CONCLUSION

In summary, this study addresses the challenge of accurately predicting student performance using two distinct datasets and a suite of machine learning techniques. By examining both classification (Dataset1) and regression (Dataset2) frameworks, we provide a comprehensive view of student outcomes. The classification dataset revealed that **Support Vector Machine (SVM)** delivered the highest accuracy (0.75), suggesting that margin-based classification is well-suited to these academic and demographic variables. Ensemble methods such as Random Forest, Gradient Boosting, and AdaBoost also performed robustly, but fell slightly behind SVM, while simpler or

more sensitive models (KNN, Decision Tree) showed lower accuracies, potentially due to overfitting or class imbalance issues.

On the regression side, the **Random Forest Regressor** achieved an $R^2$ score of 0.90, outperforming Linear Regression, Gradient Boosting Regressor, and AdaBoost Regressor. These results indicate that ensemble-based models can capture non-linear patterns in continuous academic metrics, particularly when key features—like *Absences* for classification or *Last* (prior GPA) for regression—display strong correlations with final outcomes. The synergy of well-curated feature sets and effective model selection underscores the importance of robust data preprocessing (encoding, scaling) and hyperparameter tuning.

This research contributes to the growing field of educational data mining by systematically comparing multiple algorithms on two data modalities (categorical grades vs. numeric performance). Our findings highlight how predictive accuracy can vary significantly across models depending on data characteristics. For classification tasks, handling imbalanced classes (e.g., a majority "E" grade) remains a challenge that advanced sampling techniques or class weighting might address. For regression tasks, incorporating more nuanced behavioral or psychological features could further refine predictions, enabling early interventions for struggling students.

Looking ahead, future work may delve into *longitudinal* analysis that tracks student performance over multiple semesters. This could involve real-time streaming data or textual feedback from discussion forums, integrated via sentiment analysis or advanced natural language processing. In addition, cross-institutional transfer learning might be explored to assess whether these predictive models generalize to diverse educational contexts. Overall, the consistent success of ensemble methods in both classification and regression supports the notion that robust, ensemble-based approaches—in conjunction with thoughtful feature engineering—can provide educators with more accurate, data-driven insights to support academic success.

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